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The solar origin of the „sacred triangle“ in ancient Egypt?*

von
Dušan Magdolen

Abstract

The article deals with the phenomenon of the „sacred triangle“. A hypothesis is offered showing its possible solar origin in ancient Egypt and its use by the Egyptians in the monumental architecture. In connection with the „sacred triangle“, the attention is also focused on the names of two daytime hours used in the later period of Egyptian history.

The triangle with ratio 3:4:5 of its sides is known as the „sacred triangle“. More commonly, it is known as the Pythagorean triangle. The angles opposite the right angle measure $53^{\circ} 07' 48''$ and $36^{\circ} 52' 12''$.

The alleged use of this triangle by Egyptian builders is often discussed in Egyptological literature usually in reference to the construction of right angles of buildings¹.

Despite the differences in measurement between Old Kingdom pyramids, some are shown to have ratio 3:4 between the half of the base and the height of the pyramid measurement. This half base line, the imaginery perpendicular line and the line connecting the apex of the pyramid with the mid-point of its side form a right-angled triangle² with the 3:4:5 ratio. Table 1 contains list of the pyramids indicating such proportions.

In general, the distance given horizontally for every rise of one cubit in vertical height is an ancient Egyptian measure of slope – the *seked*. In this connection, it is worth to quote from Gillings' book³ on mathematics of the ancient Egyptians:

„The *seked* of a right pyramid is the inclination of any one of the four triangular faces to the horizontal plane of its base, and is measured as so many horizontal units per one

* The article is based on one of the subjects discussed in my PhD-thesis written in 1997 at the Czech Institute of Egyptology in Prague. I would like to express my thanks to Dr. Vivienne G. Callender for her manifold help with reading the manuscript of the article as well as to Mr. Martin Isler for taking the time with reading its draft. I am indebted also to Prof. Dr. Miroslav Verner for his kindly help with publishing the article.

¹ S. Clarke/ R. Engelbach, *Ancient Egyptian Construction and Architecture*, 1990, 67; D. Arnold, *Building in Egypt. Pharaonic Stone Masonry*, 1991, 15; R. Stadelmann, *Die ägyptischen Pyramiden. Vom Ziegelbau zum Weltwunder*, 1991, 220; D. Arnold, *Lexikon der ägyptischen Baukunst*, 1994, 194; M. Verner, *Pyramidy, tajemství minulosti*, 1996, 186.

² L. Borchardt, in: *ZÄS* 31, 1893, 15.

³ R.J. Gillings, *Mathematics in the time of the pharaohs*, 1982, 212.

vertical unit rise. It is thus a measure equivalent to our modern cotangent of the angle of slope. In general, the seked of a pyramid is a kind of fraction, given as so many palms horizontally for each cubit vertically, where 7 palms equal one cubit. The Egyptian word „seked“ is thus related to our modern word „gradient“.

However, it seems likely that the pyramids are not the only monuments with the above-mentioned ratio. The royal pyramid was built as a part of a large pyramid complex surrounded by adjacent structures. According to some scholars, the ratio 3:4:5 is also found in buildings forming the pyramid complexes, together with the temples of Egyptian gods⁴. The frequency of the „Pythagorean“ ratio in Old Kingdom monuments implies a deliberate intention rather than a haphazard coincidence during building construction.

<i>King</i>	<i>Dynasty</i>	<i>Slope</i>	<i>Site</i>
Khafra ⁵	IV	53° 10'	Giza
Userkaf ⁶	V	53°	Saqqara
Neferirkara ⁷	V	53°	Abusir
Teti ⁸	VI	53° 10'	Saqqara
Pepi I ⁹	VI	53° 10'	Saqqara
Merenra ¹⁰	VI	53° 10'	Saqqara
Pepi II ¹¹	VI	53° 10'	Saqqara

Table 1. Old Kingdom royal pyramids with 53° of slope

From table 1, those Old Kingdom royal pyramids with matching 53° slope size, it is apparent that for all of the 6th dynasty pyramids which have preserved a superstructure, a proportion 3:4 between the half of the base and a height of a pyramid originally existed. The uniformity of the pyramid superstructure measurements given in table 1 is also evident in the form of the horizontal line in diagram 1¹².

⁴ J.Ph. Lauer, in: BIFAO 77, 1977, 55-78; W. Christian-Meyer, in: MDAIK 43, 1986, 195-206.
⁵ W.M.F. Petrie, The Pyramids and Temples of Gizeh, 1883, 97ff; J.Ph. Lauer, in: BIE 37, 1956, 65; J.Ph. Lauer, Observations sur les pyramides, 1960, 8 and 95; A. Fakhry, The Pyramids, 1961, 137; V. Maragioglio/ C. Rinaldi, L'Architettura delle piramidi menfite V, 1966, 50; Stadelmann, Pyramiden, 133ff; Against it J.S. Perring, The Pyramids of Gizeh II, 1982, 3; I.E.S. Edwards, The Pyramids of Egypt, 1993, 133.
⁶ Lauer, Observations, 95; Maragioglio/ Rinaldi, L'Architettura VII, 12; Stadelmann, Pyramiden, 160.
⁷ Borchardt, Neferirkere, 12; Lauer, Observations, 8 and 95; Fakhry, Pyramids, 177; Maragioglio/ Rinaldi, L'Architettura VII, pl. 11; However, the latest research gives the angle of slope 54° 30'. See M. Verner, in: MDAIK 47, 1991, 418; Verner, Pyramid, 256-262 and 382.
⁸ J.Ph. Lauer, in: BIFAO 79, 1979, 392; Stadelmann, Pyramiden, 189; R. Stadelmann, in: LÄ IV, 1251, s.v. Pyramiden.
⁹ Lauer, in: BIFAO 79, 1979, 392; Stadelmann, Pyramiden, 195; Stadelmann, in: LÄ IV, 1253.
¹⁰ Lauer, in: BIFAO 79, 1979, 392; Stadelmann, in: LÄ IV, 1253 (but see note 244).
¹¹ Jéquier, Pepi II, Vol. II, 5; Lauer, Observations, 5 and 95; Lauer, in: BIFAO 79, 1979, 392; Stadelmann, in: LÄ IV, 1253 gives 53° 13'.
¹² Such remarkable uniformity of the pyramid superstructure measurements during the 6th dynasty is evident also in case of the satellite pyramids and partly those of the Queens' pyramids. See P. Jánosi, Die Pyramidenanlagen der Königinnen, 1996, 102, 182f. and 184 Tabelle A.

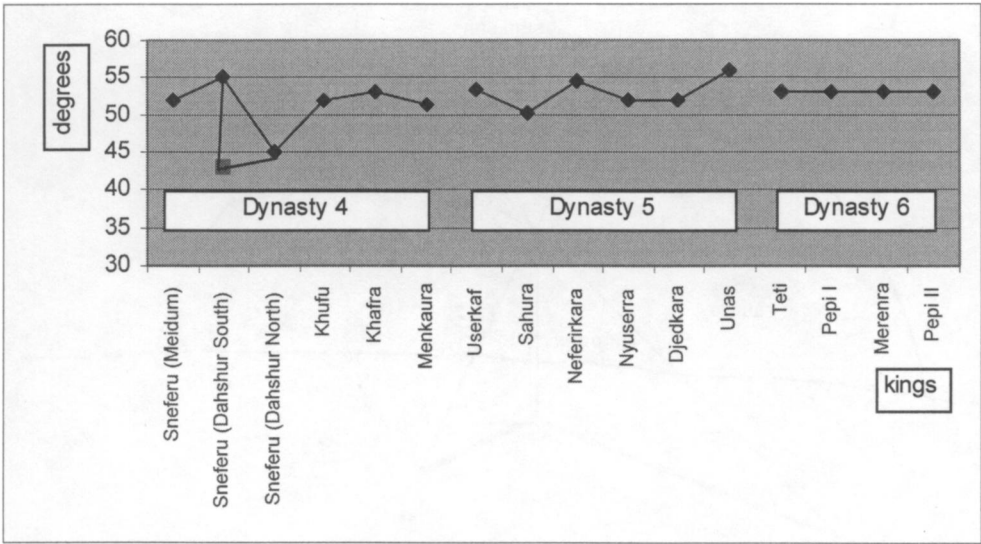


Diagram 1. Slopes of the Old Kingdom royal pyramids

There is no evidence about the discovery and use of the „sacred triangle“ by early builders, but its presence in ancient Egypt is inferred from modern surveying of extant monuments¹³. Nevertheless, let us now suppose that Egyptian pyramid builders knew of the „sacred triangle“. If it was so, they also had to know about the characteristics of such a triangle, in particular, the fact that a 90° angle was preconditional for the ratio 3:4 or 4:3. In that case, one of the questions which should be answered is – What was the origin of this special triangle, i.e. how might this ratio have been discovered by the ancient Egyptians? My suggestion, and the main hypothesis of this article, was that the triangular principle of the discussed ratio could have been derived from a long period of observing sunrays and shadows on a vertical pole (gnomon), *Solar geometry*, in fact.

Both the gnomon and its shadow form the two shorter sides of the triangle with the 90° angle at the foot of the gnomon. The sun's ray, connecting both the top of the gnomon and its shadow, forms the hypotenuse (the longest side) of such a right-angled triangle. The length of the gnomon and its shadow is both visible and measurable. The gnomon is the vertical shorter side of this triangle whose length is constant. On the other hand, the shadow is the horizontal and changeable shorter side of this triangle¹⁴ (fig.1). This results from the changing altitude of the sun. The shadow of the gnomon is fluid: it shortens in the period from sunrise until the culmination, and extends within the period from the culmination until sunset. It means that twice each day, the gnomon's shadows reach an identical length.

¹³ Cf. Arnold, Building, 15 and also Gillings, Mathematics, 238 and 242.

¹⁴ Cf. Gillings' description of the seked.

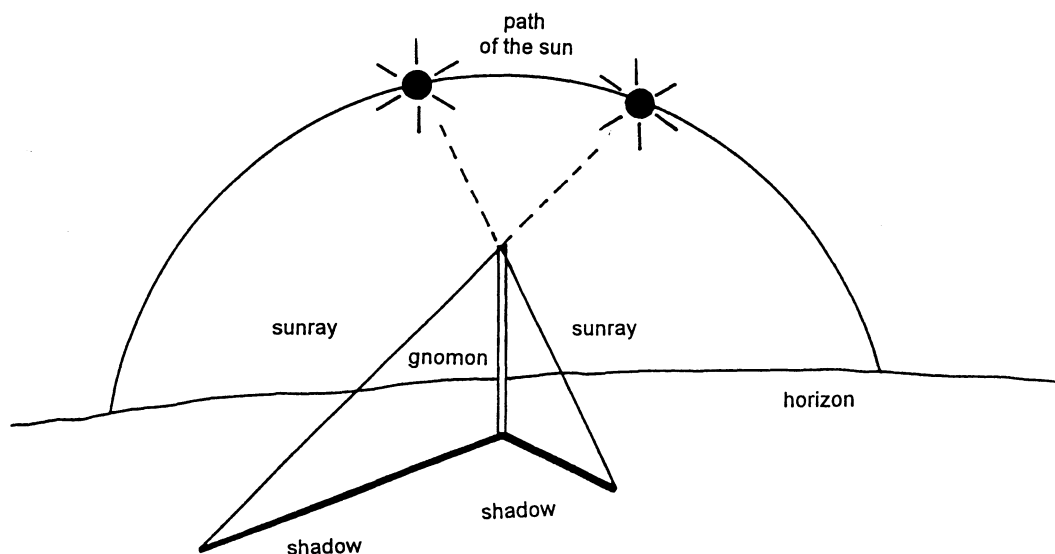


Figure 1: Gnomon, its shadows and sun's rays as the right-angled triangles

The principle of a constant vertical height and a changeable horizontal distance is very similar or even the same as in the slope of a pyramid, i.e. *škd*¹⁵. It seems that three examples from the Rhind mathematical papyrus (RMP), i.e. Nr. 57, 58 and 59 deal with the ratio 3:4. In the first case (RMP 57), the length of the side and the slope of the pyramid are known and the vertical height is calculated. In RMP 58 and RMP59, the slope of pyramid is unknown. In all three cases, the slope of the pyramid equals 5 palms and 1 finger. Such a slope corresponds with the angle¹⁶ $53^{\circ} 07' 48''$ which is the same as the angle in the „sacred triangle“.

Wolfhart Westendorf¹⁷ writes „Ägypten ist... ein Geschenk der Sonne...“. Martin Isler is of the opinion which I share, that a simple vertical pole represents an instrument used by the ancients for measuring the time or for finding the cardinal points¹⁸. Casual observation shows that the length of a shadow cast by an object in the morning, repeats itself in the afternoon, the fascinating and beautiful „shadow symmetry“ dance of the sun. It was one of our early ancestor's great successes in recognizing and understanding an aspect of nature. Now, I would like to draw attention to the possible discovery of the „sacred triangle“ in ancient Egypt.

It is well known that the inhabitable area of ancient Egypt was virtually longitudinal in character along the river, stretching from Assuan in the south to the Mediterranean Sea in

¹⁵ Wb IV, 309; R. Hannig, *Großes Handwörterbuch Ägyptisch-Deutsch*, 1995, 772.

¹⁶ L. Borchardt, in: *ZÄS* 31, 1893, 16; Gillings, *Mathematics*, 187.

¹⁷ W. Westendorf, *Altägyptische Darstellungen des Sonnenlaufes auf der abschüssigen Himmelsbahn*, *MÄS* 10, 1966, 1.

¹⁸ M. Isler, in: *JARCE* 26, 1989, 191-206; id., in: *JARCE* 28, 1991, 155-185.

the north. The large Old Kingdom royal pyramids lie between 29° and 30° of latitude north of the equator (with exception of Abu Rawash) clustering closer to 30° of latitude¹⁹. The Old Kingdom capital Memphis ($29^\circ 51'$), is also located close to this latitude, but from the south²⁰. Heliopolis ($30^\circ 08'$), the center of the sun cult, is several kilometers to the north of it²¹. On the parallel 30° of latitude (φ), the altitude of the North Celestial Pole (NCP) is of the same size, i.e. 30° and the altitude of the Celestial Equator (CE) reaches 60° above the southern horizon. Today's slope of the Earth's equator to the ecliptic is $\varepsilon = 23^\circ 27'$. The sun reaches the highest point (maximal declination) on its apparent path across the sky during the year at the the noon of the summer solstice's day when the altitude of the sun is: $h_{(s)} = (90^\circ - \varphi) + \varepsilon$. In this moment, at the 30° of latitude, the zenith distance of the sun is only $6^\circ 33'$. For the same location, the altitude of the culminating sun on the day of the winter solstice (minimal declination) is $h_{(w)} = (90^\circ - \varphi) - \varepsilon = 90^\circ - 30^\circ - 23^\circ 27' = 36^\circ 33'$. It is remarkable that the difference between this angle and the smallest one in the discussed triangle is only $19'$.

It would be interesting to know the altitude of the culminating sun on the day of the winter solstice at the same place in the earliest Egyptian history. In order to be able to calculate it, it is necessary to know the slope of the Earth's equator to the ecliptic²². Taking as an example the year 3000 BC, it was $\varepsilon_{(3000BC)} = 24^\circ 01'$, so at 30° of latitude in the winter solstice of the same year the sun would have culminated at a height of $35^\circ 59'$ above the southern horizon²³. This angles is smaller than 36° .

At $29^\circ 07'$ of latitude north of the equator, the altitude of the sun was $60^\circ 53' - 24^\circ 01' = 36^\circ 52'$.

This angle is the same as the smallest angle of the „sacred triangle“ which means that, at this area in 3000 BC during the winter solstice, the sun culminated to a height of $36^\circ 52'$ in meridian above the southern horizon. This culmination, as with every plotting of the height of the sun above the horizon, is calculated by the use of a gnomon fixed perpendicularly to the ground (2b). At the time of the winter solstice, the shadow's length from the gnomon is recorded and compared to other seasonal culminating moments of the sun, but on this day the shadow's length must have been one third longer than the proper gnomon at the moment of the culmination of the sun. Since right angle is at the foot of gnomon and the second angle is determined by the slope of the sun's ray falling down on the ground, the third angle must be $180^\circ - (90^\circ + 36^\circ 52') = 53^\circ 08'$. This angle also gives the zenith distance of the sun from the observation post. These angles are those of the so-

¹⁹ J. Baines/ J. Málek, *Atlas of Ancient Egypt*, 1992, 141.

²⁰ Baines/ Málek, *Atlas*, 135 and 234.

²¹ Baines/ Málek, *Atlas*, 135 and 234f.

²² I would like to thank to dr. J. Vondrák from the Astronomical Institute of the Czech Academy of Sciences in Prague, for data concerning the relevant positions of the sun on the 30° of latitude in ancient Egypt.

²³ J. Vondrák, e-mail from 7th November 1996.

called „sacred triangle“ formed by the gnomon (= length 3), its shadow (= length 4) and the distance connecting both the tops of gnomon and its shadow (= length 5).

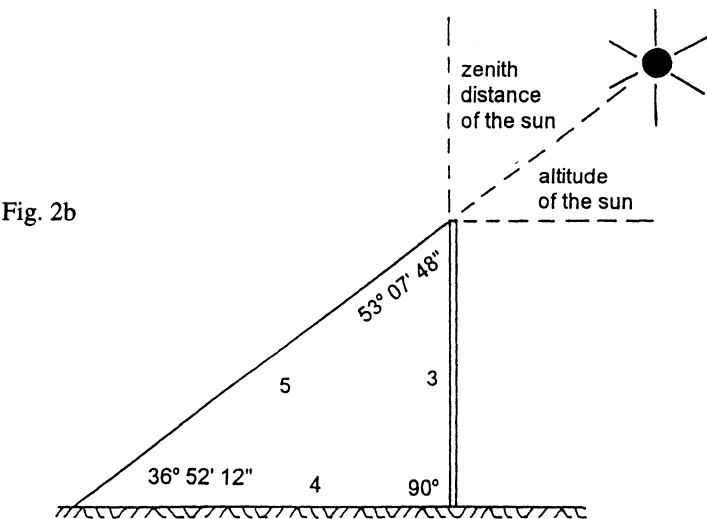
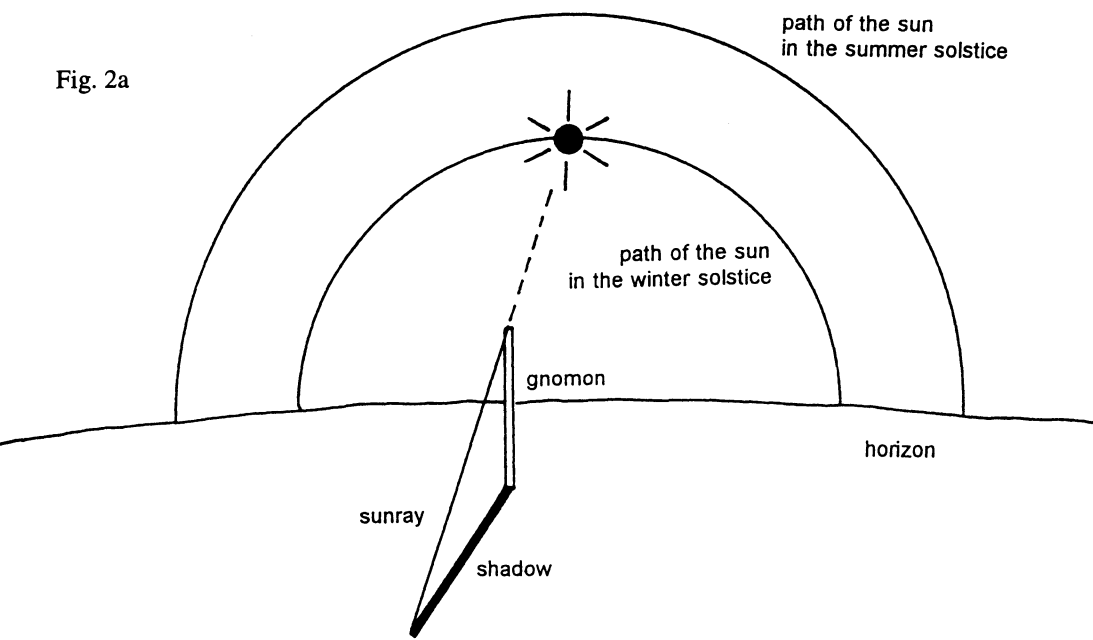


Figure 2: Gnomon, its shadow and the sunray showing the „sacred triangle“ in the winter solstice at the parallel $29^{\circ}07'$ of latitude. Figure 2a is a perspective view, 2b is a cross section

Two main factors determine the discovery of this magic ratio between the two sides of this right-angled triangle in the winter solstice: firstly, it is the ability to fix the pole

vertically to the horizontal surface in order to reach the right angle at its foot and secondly, it is the existence of a linear measure in order to be able to measure the length of the shadow. Considering these conditions related to the small deviations possibly made by the ancients, it is suggested that the discussed triangle need not to have been found exactly at the parallel $29^{\circ} 07'$, but somewhere southwards or northwards of it. In any case, from this point of view, the region north of 29° of latitude seems both theoretically and practically a strong candidate for the place where later the „sacred triangle“ might have been used. The winter solstice is a very significant day in the year, together with the summer solstice and the days of equinoxes. Observation of the length of the midday shadow day by day, would be a very good indicator of the winter solstice's day. At the same time, such observation would provide the necessary data for realizations concerning the „sacred triangle“.

Long term observations of the fluctuating lengths of shadows cast off different sizes of projectors; these resulted in several methods of measuring the time during the daylight²⁴. Several artefacts, consisting of different types of sun clocks (the earliest are attested from the New Kingdom) are preserved²⁵ from ancient Egypt. However, the earliest mention referring to the observation of the shadow and in this connection, probably, to the use of the sun clocks, dates back to the Old Kingdom²⁶. Beate George gives an example from the tomb of *Wrhww* (5th dynasty). One of his titles was . George²⁷ translates it as: „der die Elle im Schatten (oder als Schattenwerfer) herumwendet“ and in my opinion, this meaning should be interpreted likewise to the Martin Islers' hypothesis in his articles on gnomon²⁸.

The significance of actual names in hieroglyphs for the hours of daytime may also be relevant here: two are quite enigmatic, i.e. the fourth hour is named *ššt3(jt) wnw*²⁹, the secret hour³⁰ and the ninth hour is named *ḏšr ššt3 wnw*³¹ „sacred is the secret hour“(?). The fourth of twelve seasonally variable hours was the morning one before the culmination

²⁴ R.W. Sloley, in: JEA 17, 1931, 170-174.

²⁵ L. Borchardt, *Zeitmessung*, in: E. von Bassermann-Jordan, *Geschichte der Zeitmessung und der Uhren* I, 1920, 26-53; M. Clagett, *Ancient Egyptian Science II* (Calendars, Clocks, and Astronomy), 1995, 83-98 and 463-470.

²⁶ E. Graefe, in: LÄ V, 1106, n. 2, s.v. Sonnenuhr.

²⁷ B. George, in: StAeg 1, Fs. Wessetzky, 1974, 109.

²⁸ Isler, in: JARCE 28, 1991, 180f., figs. 31 and 32. But see also 156ff, figs. 1, 2 and 3; M. Isler, in: *Varia Aegyptiaca* 7/1, 1991, 67, fig. 8; However, as regards the picture (fig. 32 in JARCE and fig. 8 in *Varia Aegyptiaca*) on which a hand-held solar clocks is shown, I think, originally the plumb line was hanging from one side only. In my opinion, as is shown on that picture, the cord could be brush the engraved mark or catch it because of mass of the plumb. Then, the instrument could not have been in correct position. Therefore, I suggest, the plumb line had to hang from one side only, being unconnected with the opposite one. See, for instance, R.W. Sloley, in: JEA 17, 1931, figs. 6 and 8.

²⁹ Wb IV, 300.

³⁰ Clagett, *Science III*, 89.

³¹ Wb V, 614; For other variant of the name of this hour see Naville, *Deir el-Bahari IV*, 10, Pl. CXIV.

of the sun and the ninth hour belonged to the afternoon hours. Concerning the shadow length of the gnomon, the ninth hour corresponded to the fourth one. I was interested in the meaning of these names and tried to find an answer to the question – What kind of secret could be hidden in the names of the mentioned hours?

For the length of the daytime hour on 30° of latitude north of the equator at the summer solstice, Henne³² gives 1,16 h and it means approximately 1 h 10 min = 70 min. Tables 2a-d contain data showing positions of the sun on the day of the summer solstice. They are divided into the 12 time intervals given by the length of the ancient Egyptian hour in the summer solstice. The data are calculated for the area 30° of latitude in Egypt in 3000-1 BC. Each table consists of 12 daytime hours, the true solar time, the azimuth of the sun (measured from the north in clockwise direction) and the height of the sun above the horizon³³.

<i>Hour</i>	<i>True solar Time (h, min)</i>	<i>Azimuth (°)</i>	<i>Height (°)</i>
First	04 56	61,42	-,83
Second	06 07	69,62	13,09
Third	07 17	76,85	27,74
Fourth	08 28	83,92	42,81
Fifth	09 39	92,18	58,09
Sixth	10 49	106,61	73,19
Seventh	12 00	180,00	84,02
Eighth	13 11	253,39	73,19
Ninth	14 21	267,82	58,09
Tenth	15 32	276,08	42,81
Eleventh	16 43	283,15	27,74
Twelfth	17 53	290,38	13,09
Sunset	19 04	298,58	-,83

Table 2a: Data for year 3000 BC, (JD = 625507)

<i>Hour</i>	<i>True solar Time (h, min)</i>	<i>Azimuth (°)</i>	<i>Height (°)</i>
First	04 56	61,53	-,83
Second	06 07	69,73	13,09
Third	07 18	76,97	27,74
Fourth	08 28	84,06	42,81
Fifth	09 39	92,36	58,08
Sixth	10 49	106,92	73,16
Seventh	12 00	180,00	83,92
Eighth	13 11	253,08	73,16
Ninth	14 21	267,64	58,08
Tenth	15 32	275,94	42,81
Eleventh	16 42	283,03	27,74
Twelfth	17 53	290,27	13,74
Sunset	19 04	298,47	-,83

Table 2b: Data for year 2000 BC, (JD = 990750)

³² W. Henne, in: ZÄS 118, 1991, 27.
³³ J. Vondrák, e-mail from 7th Nov. 1996 and 20th May 1997. The Julian date is defined that JD = 2451545 corresponds with the 1st Jan. 2000 AD. The sunrise and sunset is defined for its upper margin.

<i>Hour</i>	<i>True solar Time (h, min)</i>	<i>Azimuth (°)</i>	<i>Height (°)</i>
First	04 57	61,66	-,83
Second	06 07	69,86	13,10
Third	07 18	77,11	27,74
Fourth	08 28	84,22	42,81
Fifth	09 39	92,58	58,06
Sixth	10 49	107,29	73,12
Seventh	12 00	180,00	83,81
Eighth	13 11	252,71	73,12
Ninth	14 21	267,42	58,06
Tenth	15 32	275,78	42,81
Eleventh	16 42	282,89	27,74
Twelfth	17 53	290,14	13,10
Sunset	19 03	298,34	-,83

Table 2c: Data for year 1000 BC, (JD = 1355991)

<i>Hour</i>	<i>True solar Time (h, min)</i>	<i>Azimuth (°)</i>	<i>Height (°)</i>
First	04 57	61,80	-,83
Second	06 07	70,00	13,10
Third	07 18	77,25	27,74
Fourth	08 28	84,39	42,80
Fifth	09 39	92,80	58,05
Sixth	10 49	107,67	73,08
Seventh	12 00	180,00	83,08
Eighth	13 11	252,33	73,08
Ninth	14 21	267,20	58,05
Tenth	15 32	275,61	42,80
Eleventh	16 42	282,75	27,74
Twelfth	17 53	290,00	13,10
Sunset	19 03	298,00	-,83

Table 2d: Data for year 1 BC, (JD = 1721233)

The tables show that throughout the centuries, the apparent path of the sun across the sky remained almost unchanged. But let us see the two specific hours we mentioned before. At the beginning of the fourth hour, the azimuth of the sun was 84° and its altitude was less than 43°. At the end of the same hour, the azimuth was more than 92° and the altitude was a little above 58°. What does it mean for a man working with the gnomon? Let us assume that such man located at the site we know to be 30° of latitude, was observing the shadows of a vertical gnomon and trying to measure the time. A few minutes after the beginning of the fourth hour, the shadow of the gnomon was of the same length as the gnomon itself because the altitude of the sun was 45° (ratio 1:1). The gnomon and the shadow as well as the distance between their tips formed another unique right-angled triangle – the isosceles one. Several minutes later, „the Pythagorean“ triangle would have been observed when the sun reached the altitude of 53° 07' 48", because the sun reached the same angle at the Ascendant. One would expect that this position would be simultaneous to the sun transiting

this plane. However it seems to be not the case. Based on the computer data³⁴, we can say that the sun transited the east-west plane at the following altitudes:

53° 30' for year 1 BC

53° 52' for year 1000 BC

54° 11' for year 2000BC

54° 29' for year 3000BC

It is evident that the height 53° 07' 48" of the sun above the eastern horizon was reached before transiting the east-west plane on the day of the summer solstice. The „sacred triangle“ would have been formed by the gnomon (in this case the length = 4), its shadow (the length = 3) and sun's ray connecting both the tops of the gnomon and its shadow (the length = 5). It could have been again a very good indicator of the moment for the relevant position of the sun situated just very close to the east-west plane as well as for the significant orientation of the shadow at the same moment directed to the west³⁵.

During the same day, after the upper culmination of the sun, the situation was repeated above the western horizon at the corresponding hour in the afternoon, i.e. the ninth hour. At the beginning of this hour, the azimuth of the sun was more than 267° and the altitude was more than 58°. At the end of the same hour, the sun had moved to the azimuth about 276° and the altitude was therefore more than 42°. The difference between the situations in the morning and in the afternoon consists in the converse sequence of events. Both situations are depicted in figure 3.

Therefore, the above-mentioned positions of the sun might have been of relevant significance for the names of the hours in which they occurred. Of course, it has to be realized that those events occurred only on the days around the summer solstice, and were caused by the changing daily declination of the sun. Concerning the introduction of the names for daytime hours – the date of which still remains unsolved – the situation seems to be analogous to that of the Egyptian civil calendar. The fact that the time of the summer solstice occurred around the heliacal rising of Sirius which heralded the inundation of the Nile, may have held strong association for the Egyptians. Did the „sacred triangle“ (being formed by a gnomon during the fourth and ninth hours) play any part in the announcement of the inundation?

As was demonstrated, the „sacred triangle“ might have been derived from the solar geometry of a vertical pole and its shadow. The ratio 3:4 or 4:3 was in all likelihood used for desing in the earliest of the ancient Egyptian monumental architecture. Perhaps, it was later recorded obliquely in the mathematical papyrus for problem solving, and alluded to in the occult names of the daytime hours in religious texts.

³⁴ J. Vondrák, e-mail of 23rd May 1997.

³⁵ In this connection it is interesting to notice that all the objects with such a side inclination and smaller, i.e. including the pyramids, do not cast any shadow because the sun's rays are parallel to the side which is inclined towards the sun.

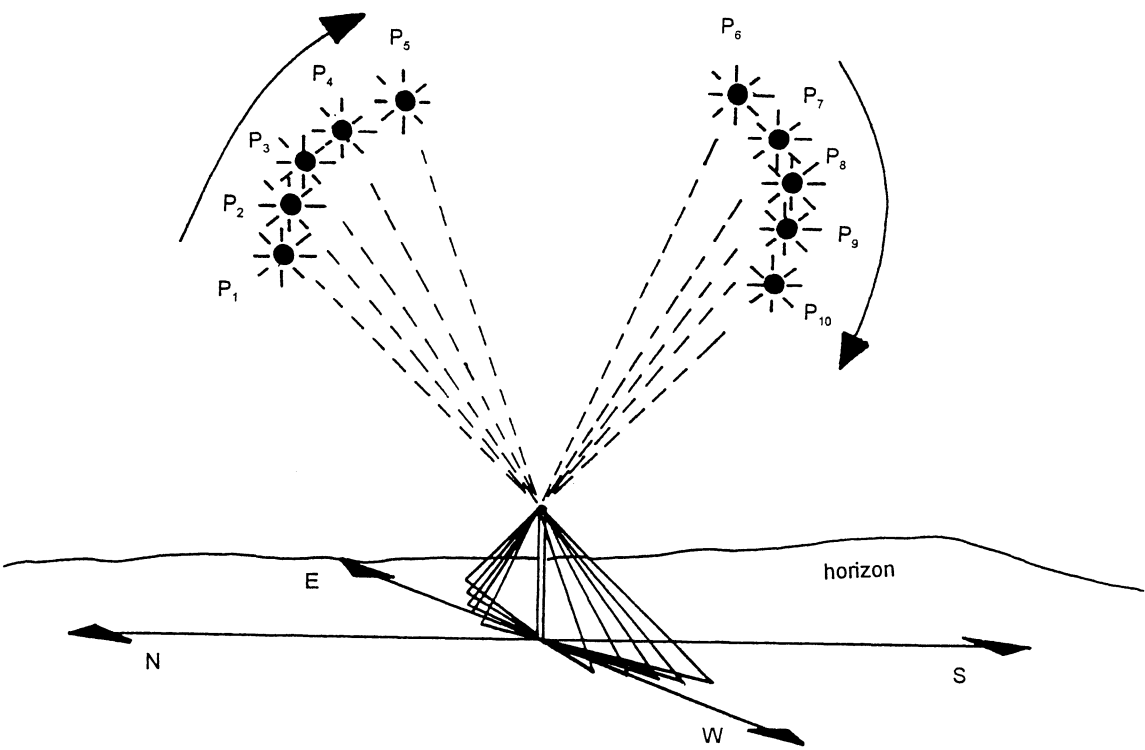


Fig. 3: Relevant positions of the sun during the fourth and ninth hours

- | | |
|--|--|
| P ₁ – beginning of the 4 th hour | P ₆ – beginning of the 9 th hour |
| P ₂ – the isosceles triangle (ratio1:1) | P ₇ – transit of the E-W plane |
| P ₃ – the „sacred triangle“ (ratio3:4) | P ₈ – the „sacred triangle“ (ratio3:4) |
| P ₄ – transit of the E-W plane | P ₉ – the isosceles triangle (ratio1:1) |
| P ₅ – end of the 4 th hour | P ₁₀ – end of the 9 th hour |

Indeed, in this writer's opinion, the so-called „sacred triangle“ could be an unsung invention first thought of by the ancient Egyptians, predating by some millenia the school of Pythagoras.